

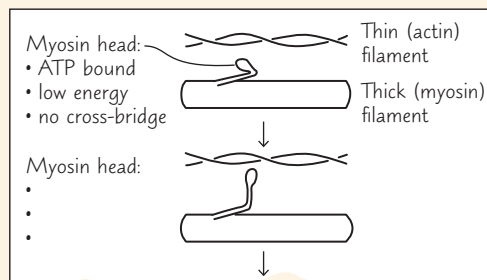
50 Sensory and Motor Mechanisms

KEY CONCEPTS

- 50.1** Sensory receptors transduce stimulus energy and transmit signals to the central nervous system p. 1108
- 50.2** In hearing and equilibrium, mechanoreceptors detect moving fluid or settling particles p. 1112
- 50.3** The diverse visual receptors of animals depend on light-absorbing pigments p. 1117
- 50.4** The senses of taste and smell rely on similar sets of sensory receptors p. 1123
- 50.5** The physical interaction of protein filaments is required for muscle function p. 1125
- 50.6** Skeletal systems transform muscle contraction into locomotion p. 1132

Study Tip

Diagram a process: To help you understand the sliding-filament mechanism of muscle contraction, fill in the remaining steps of the process. For each step, draw the position of the myosin head and list 1) the form of ATP or its components bound; 2) the energy state (low/high); and 3) whether a cross-bridge forms at that step.



Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 50
- Figure 50.18 Walkthrough: Response of a Photoreceptor Cell to Light
- BioFlix® Animation: Muscle Contraction

For Instructors to Assign (in Item Library)

- Activity: Structure and Function of the Eye
- Tutorial: Energy Costs of Locomotion



Figure 50.1 Tunneling beneath North American wetlands, the star-nosed mole (*Condylura cristata*) finds its prey in almost total darkness. To do so, the mole relies on a star-shaped cluster of appendages, each with 25,000 touch-sensitive receptors. Relaying signals to the mole's brain, the receptors initiate information processing that enables the animal to capture and eat its prey in a fraction of a second.

What steps link sensory stimuli to animal activity?

